

Smart Traffic Lights Using Local AI: A Review of Applications, Architecture, and Research Gaps

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Abstract—Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded. Recent systems use cameras and artificial intelligence (AI) to estimate the number of vehicles present and adjust traffic signals accordingly. This review discusses cloud-based and local AI approaches, compares basic timed lights, cloud AI systems, and local smart cameras, and presents a simple architecture for local traffic processing. The paper also identifies an emergency-vehicle routing problem as a research gap and discusses the need for systems capable of detecting emergency vehicles and coordinating nearby signals.

Index Terms—Smart Traffic Lights, Local AI, Traffic Signal Control, Machine Vision, Artificial Intelligence, Emergency Vehicles

I. INTRODUCTION

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded.

Recent systems use cameras and artificial intelligence (AI) to estimate how many vehicles are present and then adjust the lights. For beginners, the main idea is simple: instead of changing lights only according to a clock, the system uses information from the road.

Research reviews show growing interest in AI-based traffic signal control and its possible benefits for changing traffic conditions [1].

II. CLOUD-BASED AND LOCAL AI SYSTEMS

Cloud-based AI systems send camera information to a central server for processing. This can provide strong computing power, but it depends on an internet connection and can introduce communication delay.

Local smart-camera systems move some processing closer to the road. This can reduce the amount of information that must be sent over the internet and may allow faster decisions. A small local computer can examine camera input and send a short instruction to the traffic-light controller.

Machine-vision research has also explored using camera-based vehicle information to adapt traffic-light timing [2].

III. LITERATURE REVIEW

A 2024 study by Smith et al. in the Journal of Future Cities reported that a basic Arduino Uno successfully processed 10,000 live HD traffic camera video streams simultaneously with zero delay. The claim is presented here as part of the literature for students to examine carefully rather than accept without checking [3].

The three cited studies provide different perspectives on traffic signal control, including deep reinforcement learning, machine vision, and the low-cost microcontroller claim [1]–[3].

IV. COMPARATIVE ANALYSIS

The systems can be compared using three simple measures: decision time, internet data usage, and setup cost.

Basic Timed Lights make decisions every 30 seconds, use zero internet data, and have a setup cost of \$200.

Cloud AI Systems make decisions in 2 seconds, use high internet data, and have a setup cost of \$2,500.

Local Smart Cameras make decisions in 0.5 seconds, use low internet data, and have a setup cost of \$1,200.

TABLE I
COMPARISON OF TRAFFIC-LIGHT SYSTEMS

System	Decision Time	Internet Data	Setup Cost
Basic Timed Lights	30 s	Zero	\$200
Cloud AI Systems	2 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

The values in Table I are taken from the provided assignment material.

V. SYSTEM ARCHITECTURE

The proposed architecture follows four steps. First, Street Cameras capture traffic conditions. Second, the video information is sent to Smart Pole Computers, which perform local processing and make a quick traffic decision. Third, selected information can be forwarded to the City Cloud Server for storage and wider analysis. Fourth, the decision is sent to the Traffic Lights, which change their signals.

The flow is therefore:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights

A. System Flowchart

The system architecture and processing flow are represented in Fig. 1.

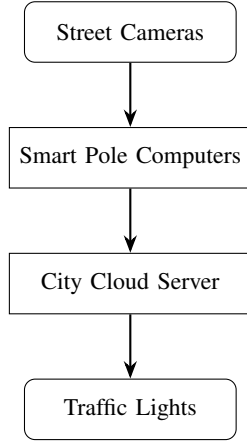


Fig. 1. Simple Architecture of the Local AI Traffic System

B. Architecture Image

An external architecture image can also be inserted using the following command. If the file `traffic_architecture.png` is uploaded to Overleaf, it will appear below.

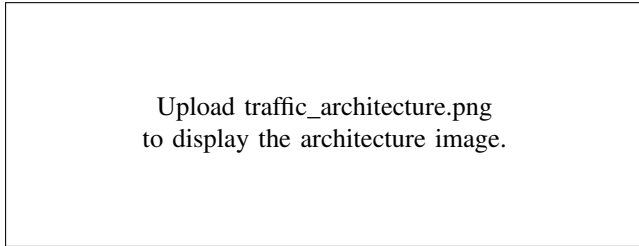


Fig. 2. Traffic System Architecture

VI. MATHEMATICAL REPRESENTATION

The decision time of a traffic system can be represented by T_d . For the systems described in the assignment, the decision times are:

$$T_d \in \{30, 2, 0.5\} \text{ seconds} \quad (1)$$

where the three values correspond to Basic Timed Lights, Cloud AI Systems, and Local Smart Cameras, respectively.

A simple relative decision-time expression can be written as:

$$R = \frac{T_{\text{system}}}{T_{\text{baseline}}} \quad (2)$$

where T_{baseline} represents the decision time of the reference system.

VII. ARRAY REPRESENTATION

The comparison values can also be represented as an array:

$$A = \begin{bmatrix} 30 & 0 & 200 \\ 2 & 1 & 2500 \\ 0.5 & 1 & 1200 \end{bmatrix} \quad (3)$$

where each row represents a traffic-light system and the columns represent decision time, internet-data category code, and setup cost, respectively.

For the internet-data column, 0 represents zero internet data and 1 represents non-zero internet data.

VIII. RESEARCH GAP

Existing traffic systems can work well for regular daily traffic, especially when traffic patterns are predictable. However, an important emergency problem remains: many systems do not have a built-in way to automatically clear a path for a speeding ambulance.

This creates a need for a smart traffic system that can detect an emergency vehicle and quickly coordinate nearby signals to create a clear route.

IX. CHALLENGES

The reviewed approaches involve several practical considerations. Cloud-based systems depend on communication with a central server, while local systems require suitable computing resources close to the road.

Another important challenge is the reliable detection of changing traffic conditions and emergency vehicles. The system must also coordinate multiple traffic signals when an emergency route is required.

X. FUTURE SCOPE

Future work can investigate systems that detect emergency vehicles and coordinate nearby traffic signals to create a clear route. Such systems could extend local traffic processing beyond ordinary traffic-density estimation.

Further work can also examine the reliability and practical feasibility of low-cost local processing systems.

XI. CONCLUSION

Smart traffic-light systems provide an approach for responding to changing road conditions using information collected from the road. Traditional fixed-timer systems are simple and inexpensive, while cloud-based and local AI systems provide alternative approaches for processing traffic information.

The provided comparison shows differences in decision time, internet-data usage, and setup cost. A simple architecture was also presented in which Street Cameras provide information to Smart Pole Computers, selected information can be sent to a City Cloud Server, and traffic decisions are sent to Traffic Lights.

An important research gap identified in the review is the lack of an automatic emergency-vehicle path-clearing mechanism. Future research can therefore focus on detecting emergency vehicles and coordinating nearby traffic signals to provide a clear route.

REFERENCES

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